SUKJU OH

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EDUCATION

Dongguk University | Seoul, South Korea
M.S. in Computer Science and Artificial Intelligence | Mar. 2026 ~ Present
- Advisor: *Prof. Sukkyu Sun*
- Selected Coursework: Mathematical Statistics (A+), Medical Image Analysis (A+)

Dongguk University | Seoul, South Korea
B.S. in Statistics & Management Information Systems (double major) | Mar. 2020 ~ Feb. 2026
- Advisor: *Prof. Yung-Seop Lee*
- GPA: **4.13 / 4.5 (Statistics major: 4.23 / 4.5)**
- Selected coursework: Machine Learning and Data Science (A), Deep Learning Applications (A), Data Structures (A+), Mathematical Statistics (A), Stochastic Processes (A+), Time Series Analysis (A+), Multivariate Analysis (A+), Regression Analysis (A), Survival Analysis (A+), Linear Programming (A), Calculus I, II (A, A+)
- Mandatory military service, Republic of Korea, Jul. 2021 ~ Jan. 2023 (leave of absence)

RESEARCH INTERESTS

- Vision-Language Models
- Medical Image Analysis
- Biomedical Signal Processing
- Surgical Robot Learning
- Efficient Sequence Modeling

PUBLICATIONS

** Corresponding author, † Equal contribution*

Journal Articles

- **Sukju Oh**, Soo Ick Cho, Dae Hun Suh, Sukkyu Sun*, "**VL-AcneSeg: A Vision-Language Framework for Region-Aware Acne Lesion Segmentation**", *IEEE Journal of Biomedical and Health Informatics*, 2026 (Early Access).
- **Sukju Oh**, Moo-Yong Rhee*, Sukkyu Sun*, "**ArrhyMamba: Explainable Unsupervised PPG Anomaly Detection for Arrhythmia via State Space Models**", *Applied Soft Computing* (Under Revision), 2026.

Conference Papers

- **Sukju Oh**, Sukkyu Sun*, "**SurgicalMamba: Dual-Path SSD with State Regramming for Online Surgical Phase Recognition**", under review at *International Conference on Learning Representations (ICLR)*, 2027.
- Hojung Lee†, **Sukju Oh**†, Sukkyu Sun*, "**Joint Video-Action Generation for Data-Scarce Surgical Robot Learning**", under review at *IEEE International Conference on Robotics and Automation (ICRA)*, 2027.
- **Sukju Oh**, Moo-Yong Rhee, Jae-Sik Jang, Sukkyu Sun*, "**Cardiologent: Multi-Agent Clinical Decision Support for Patient-Level Arrhythmia Assessment, Urgency, and Management**", under review at *AAAI Conference on Artificial Intelligence (AI for Social Impact Track)*, 2027.

PATENTS

- Sukkyu Sun, Sukju Oh, Moo-Yong Rhee, "Photoplethysmography anomaly detection using generative models", KR-Application No. 10-2025-0030999 (pending)
- Sukkyu Sun, Sukju Oh, "Method and device for assisting in the diagnosis of acne lesions", KR-Application No. 10-2025-0129404 (pending)
- Sukkyu Sun, Sukju Oh, "Method and device for generating long-term bio signals based on digital twins", KR-Application No. 10-2025-0129405 (pending)

RESEARCH EXPERIENCE

- Graduate Researcher, DGU AIM Lab, Department of Computer Science and Artificial Intelligence, Dongguk University (*Advisor: Prof. Sukkyu Sun*), Mar. 2026 ~ Present
- Undergraduate Research Intern, DGU AIM Lab, Department of Computer Science and Artificial Intelligence, Dongguk University (*Advisor: Prof. Sukkyu Sun*), Jun. 2024 ~ Feb. 2026
- Clinical collaborations: Department of Dermatology, Seoul National University Hospital; Department of Cardiology, Dongguk University Ilsan Hospital

PROJECTS

- "Photoplethysmography anomaly detection using generative models", Ministry of Education and the National Research Foundation (COSS Program), Korea, Sep. 2024 ~ Feb. 2025

AWARDS AND SCHOLARSHIPS

- Research Assistant Scholarship, Dongguk University, Fall 2024 – Spring 2025

- Teaching Assistant Scholarship, “Exploratory Data Analysis”, Dongguk University, Fall 2024 – Spring 2026

SKILLS

- Programming: *Python, PyTorch, R, C*
- Languages: *Korean (native), English*